DATASET INSPECTION REPORT

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Internship Programme
AnalystLab Africa

Week 1 Assignment

Prepared by: Christian Agyapong

Table of Contents

Table of Contents

1. Dataset Overview

The success of any Machine Learning project depends heavily on the quality and structure of the dataset used for model development. Before training a predictive model, it is important to understand the dataset, identify potential quality issues, and determine the preprocessing steps required to prepare the data for analysis.

For this project, the **Telco Customer Churn Dataset** was selected because it represents a real-world telecommunications customer database containing demographic information, subscribed services, account details, billing information, and customer churn status.

The dataset will be used to develop a Machine Learning model capable of predicting whether a customer is likely to discontinue the company's services.

2. Dataset Source

- **Dataset Name:** Telco Customer Churn Dataset
- **Source:** Kaggle
- **File Format:** CSV (Comma-Separated Values)

Dataset Summary

Attribute	***Description***
Total Records	7,043 customers
Total Features	21 columns
Target Variable	Churn
Problem Type	Binary Classification
File Type	CSV

The dataset contains one record for each customer together with demographic information, subscribed services, billing details, and whether the customer eventually churned.

3. Dataset Structure

The dataset consists of twenty-one variables describing customer characteristics and one target variable indicating customer churn.

The variables can be grouped into the following categories:

Customer Identification

- customerID

Demographic Information

- gender
- SeniorCitizen
- Partner
- Dependents

Account Information

- tenure
- Contract
- PaperlessBilling
- PaymentMethod

Services Subscribed

- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies

Financial Information

- MonthlyCharges
- TotalCharges

Target Variable

- Churn

Grouping the variables in this way provides a clearer understanding of the different aspects of customer behaviour represented within the dataset.

4. Data Types

The dataset contains both numerical and categorical variables.

The table below summarizes the data types.

Feature	Data Type	Description
customerID	Object	Unique customer identifier
gender	Object	Customer gender
SeniorCitizen	Integer	Senior citizen indicator (0 or 1)
Partner	Object	Whether customer has a partner
Dependents	Object	Whether customer has dependents
tenure	Integer	Number of months with the company
PhoneService	Object	Phone service subscription
MultipleLines	Object	Multiple phone lines
InternetService	Object	Type of internet service
OnlineSecurity	Object	Online security service
OnlineBackup	Object	Online backup service
DeviceProtection	Object	Device protection service

TechSupport	Object	Technical support service
StreamingTV	Object	Streaming television service
StreamingMovies	Object	Streaming movies service
Contract	Object	Contract type
PaperlessBilling	Object	Paperless billing status
PaymentMethod	Object	Payment method
MonthlyCharges	Float	Monthly customer payment
TotalCharges	Object*	Total amount paid by customer
Churn	Object	Customer churn status

Observation

Although **TotalCharges** represents a numerical value, it is stored as an object because some records contain blank strings. This column will require conversion to a numeric data type during preprocessing.

5. Missing Values Analysis

An essential step in data inspection is identifying missing or incomplete information that may affect model performance.

Using the following command:

df.isnull().sum()

the dataset initially appears to contain no missing values.

However, a closer inspection reveals that the **TotalCharges** column contains blank spaces (" ") instead of standard missing values (NaN).

These blank entries prevent the column from being recognised as a numerical feature.

To correctly identify these missing values, blank strings should first be replaced with NaN before converting the column to a numeric data type.

Observation

- Most variables contain complete information.
- TotalCharges contains blank entries that require cleaning.
- Missing values must be handled before model development.

6. Duplicate Records Analysis

Duplicate observations can introduce bias during model training by overrepresenting certain customers.

The dataset should therefore be inspected using:

```
df.duplicated().sum()
```

Observation

No duplicate customer records were identified within the dataset.

Each customer appears only once, indicating that the dataset is suitable for predictive modelling without duplicate removal.

7. Feature Descriptions

The table below describes the purpose of each variable.

Feature	Description
customerID	Unique identifier assigned to each customer
gender	Customer gender
SeniorCitizen	Indicates whether the customer is a senior citizen
Partner	Indicates whether the customer has a partner

Dependents	Indicates whether the customer has dependents
tenure	Number of months the customer has remained with the company
PhoneService	Indicates whether the customer subscribes to phone service
MultipleLines	Indicates whether multiple phone lines are subscribed
InternetService	Type of internet connection
OnlineSecurity	Online security subscription
OnlineBackup	Online backup subscription
DeviceProtection	Device protection subscription
TechSupport	Technical support subscription
StreamingTV	Streaming television subscription
StreamingMovies	Streaming movie subscription
Contract	Type of customer contract
PaperlessBilling	Paperless billing preference
PaymentMethod	Method used to make payments
MonthlyCharges	Monthly subscription payment
TotalCharges	Total amount paid by the customer
Churn	Indicates whether the customer left the company

Understanding each feature helps identify variables that are likely to contribute to customer churn prediction.

8. Data Quality Issues

Several issues were identified during the inspection process.

1. Incorrect Data Type

The **TotalCharges** feature is stored as an object instead of a numerical variable due to blank entries.

2. Missing Values Represented as Blank Strings

Blank strings should be converted into proper missing values (NaN) before further analysis.

3. Presence of Categorical Variables

Most features are categorical and cannot be used directly by Machine Learning algorithms. They must be converted into numerical representations using encoding techniques.

4. Identifier Column

The **customerID** feature uniquely identifies each customer but provides no predictive value. It should therefore be removed before model training.

5. Feature Scaling

Algorithms such as Support Vector Machine (SVM) and K-Nearest Neighbours (KNN) perform better when numerical variables are scaled. Features such as **tenure**, **MonthlyCharges**, and **TotalCharges** should be standardised or normalised where appropriate.

Overall, the dataset is considered to be of good quality with only minor preprocessing requirements.

9. Preprocessing Requirements

Before training a Machine Learning model, the following preprocessing steps should be completed.

Step 1

Remove the **customerID** column because it does not contribute to prediction.

Step 2

Replace blank strings in **TotalCharges** with missing values (NaN).

Step 3

Convert **TotalCharges** from an object data type to a numerical data type.

Step 4

Handle missing values using an appropriate imputation strategy, such as replacing missing values with the median.

Step 5

Encode categorical variables using One-Hot Encoding or Label Encoding, depending on the modelling approach.

Step 6

Scale numerical variables where required to ensure algorithms that rely on feature magnitude perform effectively.

Step 7

Separate the dataset into input features (**X**) and the target variable (**y**) in preparation for model training.

These preprocessing steps will improve data quality and ensure compatibility with Machine Learning algorithms.

10. Conclusion

The dataset inspection process provided a comprehensive understanding of the Telco Customer Churn dataset used in this project.

The dataset contains **7,043 customer records** and **21 variables**, representing customer demographics, subscribed services, account information, billing details, and churn status. The inspection identified that the dataset is generally clean, with no duplicate records and only minor issues requiring preprocessing.

The primary data quality concern involves the **TotalCharges** variable, which contains blank entries and is stored as an object rather than a numerical data type. In addition, categorical variables will require encoding before they can be used for Machine Learning.

Overall, the dataset is suitable for customer churn prediction after the recommended preprocessing steps have been completed. The cleaned dataset will provide a strong foundation for exploratory data analysis, feature engineering, and the development of predictive Machine Learning models in the subsequent stages of the project.